

SPECIAL ISSUE ARTICLE **OPEN ACCESS**

Strategic Human Resource Management in the Era of Algorithmic Technologies: Key Insights and Future Research Agenda

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Received: 8 November 2024 | **Accepted:** 11 November 2024

Funding: This work was supported by Marcus Wallenbergs Stiftelse för Internationellt Vetenskapligt Samarbete.

Keywords: algorithmic management | artificial intelligence | HR and technology | HRIS | information systems | strategic HR | technology

ABSTRACT

This article presents a contemporary review of human resource management (HRM) research on algorithmic technologies, including artificial intelligence, machine learning, and natural language processing. By connecting these recent advancements to the long-standing scholarly tradition of HRM-technology relations, this review examines current knowledge on how algorithmic technologies are reshaping three key areas: (1) work structures and design, (2) HR delivery activities, and (3) the management of technology workers. Using a threefold conceptualization of technology—the tool view, proxy view, and ensemble view—this review explores how organizations employ algorithmic systems to enhance productivity, how the human agency interacts with and resists these technologies, and how broader social, cultural, and institutional contexts shape the use of algorithms in HRM. Additionally, this article offers suggestions for future research, highlighting the unique opportunities algorithmic technologies provide to HR scholars for making enduring contributions to the broader conversations on HRM and technology.

1 | Introduction

The use of algorithmic technologies in human resource (HR) management is changing the landscape of organizational operations. Artificial intelligence (AI), machine learning (ML), natural language processing (NLP), and large language model technologies have become increasingly important in managing vast amounts of employee data, streamlining HR decision-making processes, and improving the efficiency of HR activities. While the use of technology in HR management (HRM) is not new, the recent rise of algorithmic technologies represents a substantial shift from previous technical innovations. Unlike prior tools that were largely focused on administrative efficiency, algorithmic technologies have the ability to perform sophisticated cognitive tasks, significantly altering how HRM

is understood and implemented (Cheng and Hackett 2021; Giermndl et al. 2022; Strohmeier 2020).

Algorithmic technologies in HR have attracted much scholarly attention in recent years. The literature has grown so quickly that even multiple reviews have already been published on the HR implications of specific applications of algorithmic technologies (see Table 1 for a summary of recent reviews). Unfortunately, however, the development of algorithmic technologies has moved faster than scholarly research, and the relevance of existing reviews has quickly deteriorated. For instance, most reviews could not include the research on generative AI that was made widely available just a few years ago. This suggests the need for an updated and current review of HRM scholarship on algorithmic technology. More importantly, it also suggests that we need a

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review that could guide future scholars to conduct research that will make lasting impacts on the literature, rather than write papers that will soon be forgotten with the passing of a particular technology. With these needs in mind, this article seeks to draw a contemporary map of algorithmic HRM research, employing a rather distinctive approach: linking the growing research on the topic to the larger, long-standing scholarly tradition of exploring the interface of technology and HRM.

Positioning algorithmic HRM within the historical context of HR-technology research has several advantages. First, it allows us to highlight that many scholarly conversations on these technological advances are not entirely novel, rather the continuation of decades-long investigations into the role of technology in the workplace. For more than 60 years, HR scholars have made substantial efforts to make sense of the impact of technology on HRM. Back in the 1960s, for instance, HR scholars were already concerned about the impact of automation on job displacement (Blum 1964), the changes in HR functions brought about by digital data processing technologies (Austin 1965), and the challenges of managing technology workers (Broadwell 1965). Therefore, HR scholars' enthusiasm for the impact of algorithmic technologies is not a new phenomenon but an extension of the 60-plus years of collective effort. Recognizing this will help us better assess our achievements in studying algorithmic technologies, and better identify the areas in need of influential future research. Also, it allows us to appreciate the unique challenges and opportunities brought by the features of the algorithmic technologies, such as their adaptive and self-learning abilities, which sets them apart from prior technical achievements.

In this review, and with the articles published in this special issue, we advance HRM-technology literature by highlighting the connection between contemporary algorithmic HR research and the long tradition of HR scholarship on technology. To this end, we build on Kim, Wang, and Boon (2021), which identified the enduring patterns of HR research on technological development over the last 60 years. They showed that whenever new innovative technologies were made available, HR scholars attended to them from nine different angles, across three major themes (new work structure and design, new HR delivery techniques, and new technology workers), and using three assumptions on technology (tool view, proxy view, and ensemble view). Following this framework, we will survey the current status of HR scholarship on algorithmic technologies, indicating that algorithmic technology is not an entirely new phenomenon but another technological artifact that has a significant impact on HRM. In reviewing the current literature on algorithmic HR, we also highlight the unique contributions it makes to the broader research stream on HRM and technology. Then, we discuss promising avenues for future research, where algorithmic technologies provide HR scholars with the opportunity to make significant and enduring contributions to the literature.

2 | Algorithmic Technology as the Subject of HR Research

Algorithmic technologies represent a collection of adaptive approaches, including AI and ML, designed to analyze and interpret large datasets, performing cognitive tasks typically

managed by humans (Giermindl et al. 2022). Drawing on previous studies (Cheng and Hackett 2021; Strohmeier 2020), we define algorithmic technologies as “a set of computer-programmed steps to automatically accomplish a task by transforming data into output.” In other words, these technologies process data to produce “information” that can be considered new knowledge or otherwise (Pachidi et al. 2021).

Most algorithmic technologies are static, with computer-programmed steps that do not change regardless of the data input. However, newer technologies have become more adaptive and self-learning, enabling computers to perform tasks typically requiring human cognition and intervention. For instance, in AI, algorithmic technologies evolve with the data they process, due to ML capabilities (Vrontis et al. 2021).

Our interactions with algorithmic technology share many similarities with past technological innovations. Previously, tools like Microsoft Excel or IBM SPSS were used to rank job applicants or predict turnover. Today, self-learning algorithms can dynamically adjust the variables or steps needed to predict a candidate's performance within an organization (Meijerink et al. 2021). Algorithmic technologies now support or automate HR functions, such as filtering job applicants (Leicht-Deobald et al. 2019), and enhancing employee morale through detailed people analytics (Gal, Jensen, and Stein 2020).

However, algorithmic technology also has unique characteristics that differentiate it from earlier innovations. The key feature is the precision of each algorithmic step, where outcomes are uniquely defined and depend on the input and results from previous steps. The literature has highlighted several distinctive features related to HR research, including opacity (Faraj, Pachidi, and Sayegh 2018; Heiland 2022; Jarrahi et al. 2020), potential biases within systems (Langer and König 2023; Vassilopoulou et al. 2024), user preference for human over algorithmic decision-makers (Lacroux and Martin-Lacroux 2022; Mahmud et al. 2022), issues of trust in human-like algorithms (Aloisi and De Stefano 2022; De Stefano 2020; Salles, Evers, and Farisco 2020), algorithms as alternative governance mechanisms (Danaher et al. 2017; Latzer and Festic 2019), and their decision-making capabilities (Newman, Fast, and Harmon 2020).

Reviewing the HR implications of algorithmic technologies, we examined existing reviews on the theme (see Table 1 for a summary of recent reviews). While some reviews broadly explore the intersection of these technologies and HRM (Basu et al. 2023; Budhwar et al. 2022; Malik, Budhwar, and Kazmi 2023; Pan and Froese 2023), others focus more narrowly on specific technologies and/or specific HR functions (Cheng and Hackett 2021; Hesselbarth, Alfes, and Festing 2024; Garcia-Arroyo and Osca 2021; Margherita 2022; Pereira et al. 2023). These reviews consistently depict algorithmic technologies as an unprecedented new phenomenon, and highlight that HRM scholarship on this topic is in its early stage, lacking theoretical and methodological rigor. In doing so, these reviews rarely appreciate the connection between the studies on algorithmic HRM and broader research on HRM-technology relations. Many questions asked by the reviewed articles were closely related to those earlier HR scholars had asked when they encountered technological innovations of their generation, such as mainframe computers in the 1960s, HR

TABLE 1 | A summary of previous reviews on algorithmic HRM.

Authors, year, journal	Review theme and scope	Major findings
Basu et al. 2023, HRMR	• Systematic review at the intersection of AI and HRM • Fuzzy set analysis of the 100 articles	• Existing studies tend to report more on AI's beneficial organizational outcomes (such as cost reduction, employee engagement) than employees' negative reactions (such as employee resistance). • Conceptual research and empirical research tend to focus on different themes. • AI-HRM literature evolves over time.
Budhwar et al. 2022, IJHRM	• Systematic review on AI and HRM • 70 papers	• Research on the role of AI in HRM is still limited and fragmented. • Further research is needed that analyzes the role of AI-assisted applications in HRM functions, and human-AI interactions in large multinational enterprises diffusing such innovations.
Cheng and Hackett 2021, HRMR	• Integrative review of algorithms in HRM • 22 academic papers, and 122 practitioner-oriented items	• Evidence for research-practitioner divide. • The "black box" label for AI is appropriate only to a limited extent. HRM-related algorithms are better characterized as "glass boxes," as they reflect some components of a theory. • Theory-driven research is lacking. Most research is either descriptive or predictive and cannot be classified as theory driven.
Hunkenschroer and Luetge 2022, JBE	• Systematic review on ethicality of AI-enabled recruiting • 51 articles	• Ethical opportunities (e.g., reduction of human bias, process consistency, timely feedback, efficiency). • Ethical risks (e.g., algorithmic bias, privacy, lack of explainability). • Ethical ambiguities (e.g., effect on workforce diversity, informed consent, perceived fairness).
Hesselbarth, Alfes, and Festing 2024, IJHRM	• Systematic literature review on technology-driven work arrangements • 191 studies	• Complexity theory as the overarching framework for the review. • Technology-driven work arrangements are defined as flexible approaches to work within and beyond organizational boundaries. • Themes are clustered on three dimensions: structure (28 studies), process (32 studies), and actor (44 studies).
Garcia-Arroyo and Osca 2021, IJHRM	• Systematic review on Big Data contributions to HRM • 41 articles	• No sophisticated conceptualization about the role of Big Data. • Studies have provided information on the application of Big Data approaches by HR departments, especially in recruitment and selection, management of knowledge and talent, organizational behavior, health and well-being at work, and organizational efficiency.
Malik, Budhwar, and Kazmi 2023, HRMR	• Systematic review on the effect of AI on HRM • 67 articles	• Propose a "strategic framework" that guides HR managers in adopting and implementing AI in HRM. • More research is needed on ethical dimensions, employees' experiences, and the multilevel nature of the phenomenon.
Margherita 2022, HRMR	• Systematic review on HR analytics • 68 articles	• 106 key concepts were reduced to three main themes: enablers (technology factors, organizational enablers), applications (descriptive and prescriptive), and value (to employee and organization). • A conceptual framework of 'exponential HR analytics'. • The concept of AI needs further clarification.
Pan and Froese 2023, HRMR	• Systematic interdisciplinary review of AI in HRM • 184 articles	• Lack of interdisciplinary synthesis in the AI-HRM research. Engineering researchers focus on developing AI tools, while management papers focus on AI usage. • Fewer than a quarter of the studies used solid theories, and issues in methodological rigor.
Pereira et al. 2023, HRMR	• Integrative review of AI's impact on workplace outcomes • 60 papers in 30 leading journals since 1995	• Articles are grouped into different HR functions: HR planning ($n = 10$), training ($n = 25$), health and well-being ($n = 10$), performance management ($n = 7$), labor relations ($n = 5$), and recruitment/selection ($n = 4$).

information systems in the 1980s, email communications in the 1990s, and mobile devices in the 2000s. Therefore, we aimed to conduct a review that not only updates our knowledge on the topic, but also connects the study of algorithmic technologies with the broader effort in HR research to understand the implications of technology on HRM. This is a valuable endeavor as it will help us better appreciate the enduring contributions of algorithmic HRM research to the literature, beyond its capacity to explain the phenomena specific to a particular time and specific technological artifacts.

In the following sections, we identify three thematic streams at the intersection of algorithmic technologies and HRM: (1) the impact of algorithmic management on work structure and design; (2) the use of these technologies to enhance HR delivery; and (3) the emergence of new types of workers within algorithmic management. These three themes directly correspond to the major themes Kim, Wang, and Boon (2021) identified as those HR scholars have revisited whenever a new technological innovation began to affect HR practices. We also propose how these themes can be explored using the threefold conceptualization of technology—the tool view, the proxy view, and the ensemble view—relevant to HRM (Kim, Wang, and Boon 2021). The tool view understands technology as a stable and determined set of equipment, procedures, and techniques there to serve its owner as intended by its designer. It corresponds to the most traditional view of technology. The proxy view recognizes the agency of users who can not only accept the technology as designed, but also have the capability to resist and twist its usage. The proxy view was found useful to understand why an organization finds

it difficult to adopt and implement a technology. The ensemble view emphasizes the social contexts that shape and change the meaning and usage of technological artifacts. According to this view, technology is a socially constructed phenomenon, rather than a simple productivity tool. It allows researchers to appreciate complex social dynamics around technologies, and the role of institutional and cultural environments in the emergence and utilization of a technology.

Mapping out existing contributions to algorithmic HRM along the ninefold framework (three recurring themes $\times$ three technology conceptualizations), we will show how the tradition of technology-HRM research has continued in the era of algorithmic management, and what novel insights algorithmic HRM provides to the bigger scheme of the scholarly project. Also, based on our matrix of thematic streams and conceptualizations, we suggest a set of general research questions to guide future HR scholars exploring the implications of new technological artifacts (see Table 2).

3 | Theme 1: Impact of Algorithmic Management on Work Structure and Design

3.1 | Tool View—Algorithmic Technology as a Productivity Enhancement Tool

When new technologies are made available, organizations utilize them as a tool to enhance their productivity and profitability. With the new technology, organizations attempt to reduce

TABLE 2 | Future research avenue.

	Work structure and design	Algorithmic HR technologies	New types of tech workers (platform workers and algorithmic jobs)
Tool view of technology	• How does algorithmic technology change work structure and design, and thereby enhance a firm's outcomes? • What pattern of collaboration between algorithms and humans would produce the best outcomes for the company?	• How does algorithmic HR technology enhance the quality of HR decisions and/or the efficiency of traditional HR activities? • How does algorithmic technology engender new HR activities?	• How do algorithm workers (curators, brokers, articulators) contribute to the success of an organization? • How can organizations systematically hire, motivate, retrain, reward, and retain high-quality platform workers?
Proxy view of technology	• What are the sources of algorithm aversion? • What drives algoactivism? • How do humans build trust in algorithms?	• Why do some workers react differently than others to algorithmic HR technologies? • How does algorithmic HR technology change the status and identity of HR professionals?	• How do algorithm workers (curators, brokers, articulators) formulate their professional identity? • What drives the satisfaction and commitment of platform workers in various life domains?
Ensemble view of technology	• How is algorithm aversion affected by the cultural and institutional environment? • How is algorithm transparency secured by the norms and regulations in a society?	• How do social norms regarding AI affect the effectiveness of algorithmic HR technologies? • How do new regulations on AI influence the development and usage of algorithmic technologies?	• How is the career of algorithm workers shaped by cultural and institutional environments? • How do migrant platform workers make sense of their work?

costs and enhance productivity by replacing some tasks, jobs, or occupational groups represented by humans. When algorithmic technology emerged, many scholars predicted it would lead to the displacement by machines of many jobs that were once considered non-replaceable. Basu et al. (2023), in their review of the literature, observed that the loss of jobs due to digital technology was a major source of apprehension among scholars. Evidence suggests that algorithmic technology did indeed result in some job displacement. For instance, Acemoglu et al. (2022) examined online job vacancies in the United States since 2010. They reported that in organizations with many tasks compatible with AI's capabilities, companies reduced hiring in non-AI positions, while adjusting the skill requirements for remaining positions.

Although the job displacement effect of technology is not in itself a new phenomenon, algorithmic technology is different from previous technologies in terms of the nature of tasks and jobs being replaced. Unlike previous technologies that primarily affected the physical components of a job, algorithmic technologies possess capabilities for complex decision-making. For instance, algorithmic technology and AI have demonstrated their capability to replace human drivers, making many decisions that may directly affect the safety of humans (Agrawal, Gans, and Goldfarb 2017). Algorithmic technology's capability to replace cognitively complex jobs influences job seekers' attitudes toward their career prospects. Hanna et al. (2024) examined how vocational interests are being affected by new digital technologies. They suggested that investigative (scientific) and conventional (detail-oriented) vocational interests were most susceptible to automation, whereas social (people-oriented) and realistic (hands-on) vocational interests were least vulnerable.

Another distinctive feature of algorithmic technologies is their potential to become collaborative partners with human workers, rather than replace them. Jia et al. (2024) examined how algorithmic systems assisted human workers in achieving better performance. They found that AI assistant systems could effectively enhance productivity through "sequential" work design, where AI and humans divide the work sequentially so that AI handles earlier-stage repetitive tasks and humans take on the later-stage creative work. The authors' field experiment in a telemarketing company showed that algorithmic systems could enhance employees' creativity, especially those with high-level skills. Shaikh et al. (2023) conducted a study in the healthcare sector, and found that AI had a positive impact on productivity by enhancing knowledge sharing and employee well-being. Wang, Gao, and Agarwal (2024) examined how knowledge workers can team with AI to enhance overall performance. They conducted a field study in which knowledge workers acted with an AI solution for medical chart coding. They found that different knowledge workers accrued varying degrees of benefit from AI. It was more advantageous for staff with extensive task-based experience, but senior employees benefited less than their junior counterparts.

3.2 | Proxy View—Workers' Agentic Responses to Algorithmic Technology

When a new technology is introduced to an organization, humans do not accept it passively. Rather, they demonstrate

their agency, proactively make sense of it, and resist if needed. Researchers recognize that algorithmic technology is another that is subject to the process of user acceptance. According to Basu et al. (2023), nearly half of the publications on AI–HRM relations dealt with people's reactions toward algorithmic technology.

Many scholars realized that employees' reactions to algorithmic technologies could be explained as earlier studies had explained their reactions to other new technological innovations (e.g., Del Giudice et al. 2023; Mogaji et al. 2024). A theoretical framework often found useful in this respect is the technology acceptance model (TAM). TAM is a key theoretical lens in explaining why certain individuals are more (less) likely to use a new technology such as a personal computer. This perspective posits that individuals may develop varied perceptions of new technology, and their acceptance is likely when they perceive it as very useful and easy-to-use. Employing this theoretical framework, scholars showed that, as with previous technologies, algorithmic technologies would indeed be better received when seen as useful and easy-to-use. For instance, Brachten, Kissmer, and Stieglitz (2021) examined employees' acceptance and willingness to use chatbots. They found that employees' intention to use the company's chatbots was influenced more by their intrinsic motivations than external influences. They indicated the importance of employees' subjective convictions about the usefulness of the algorithmic technology, which is largely in line with previous findings on employees' acceptance of new technology.

While algorithmic technology is shown to be another case of new technology accepted through users' perceptual reactions, scholars noted that it has several distinctive features that cannot readily be captured by earlier TAMs. One of those unique features is captured by the notion of algorithm aversion. **Algorithm aversion** refers to users' unwillingness to accept and utilize advice and instructions from algorithms, and their tendency to rely on humans instead of algorithms (Dietvorst, Simmons, and Massey 2015). Unlike previous technologies that replaced some of the physical tasks that humans performed, algorithms take on the role of advisers and decision-makers, which once were considered tasks only humans could execute. In their seminal work on algorithmic aversion, Dietvorst, Simmons, and Massey (2015) showed that people were aversive to relying on algorithms, even after they saw they could outperform humans. Although the wide generalizability of algorithmic aversion is debatable (Logg, Minson, and Moore 2019), a number of studies confirmed that algorithm aversion is readily observed across many contexts (Burton, Stein, and Jensen 2020), and especially among those users with a high level of expertise (Allen and Choudhury 2022).

One of the sources of algorithmic aversion is the lack of employees' trust. The importance of "trust" in the process of adopting a new technology was discussed in cases of previous technological innovations. For instance, in the early 2000s, DeRosa et al. (2004) argued that the success of the technologically mediated virtual team depended on employees' trust. However, compared to the cases of previous technologies, trust appears to have bigger implications concerning algorithmic technology, due to the technology's inherent characteristics, that is, the lack of transparency and understandability of algorithmic systems'

underlying logic and operating mechanisms. Empirical studies have confirmed that user acceptance of algorithmic technologies is indeed affected by the user's trust in the technologies (e.g., Choung, David, and Ross 2022), and in the management who operate the algorithmic system (Weibel, Schafheitle, and van der Werff 2023).

Scholars seek to understand how “trust” works in the context of algorithmic technology. They suggest that people recognize algorithms' human-like qualities, but find it difficult to trust algorithmic systems as much as they do other humans. Turel and Kalhan (2023) proposed that people could develop an implicit bias against AI systems as threatening species. Jago (2019) observed that people believe AI algorithms are not as authentic as humans, and, therefore, their decisions were not considered to have the same moral qualities as human decisions. Glikson and Woolley (2020) delved into humans' trust in AI systems. They argued that the more intelligent the AI system, the harder it is for humans to trust it. They also proposed that AI's tangibility (capability to be touched and perceived), transparency (understandability of the inner logics and underlying operating rules), reliability (consistency over time), immediacy behaviors (socially oriented gestures), and anthropomorphism (human-likeness) are crucial for developing trust in AI.

Another theme that sets algorithmic technology research apart from that on previous technology adoptions is “**algoactivism**” (Kellogg, Valentine, and Christin 2020), which refers to the tactics and activities undertaken by individuals and collectives to resist algorithmic control. Tarafdar, Page, and Marabelli (2023) observed that algorithmic work—algorithms taking on the role of a human's co-worker—might suffer from “broken loop learning” due to humans' intentional resistance to following instructions from the algorithm. Cameron and Rahman (2021) examined the lived experiences of gig workers, focusing on workers' resistance to algorithmic control systems. They found that workers had the greatest flexibility to employ resistance at the beginning of the labor process, but had to adapt their tactics as their capacity to resist diminished with each successive stage of the process. Rahman (2021) noted that opaque algorithmic management systems created an “invisible cage” for workers, where the criteria for success and any changes to those criteria were unpredictable. He found that the opaque ranking systems in a professional service platform instigated some gig workers' experimentation with different ways to achieve higher performance scores (“experimental reactivity”). On the contrary, Rahman concluded that other gig workers reduced their dependence on the platform (“constrained reactivity”).

3.3 | Ensemble View—Algorithmic Technology in Complex Social Contexts

The ensemble perspective on technology emphasizes the significance of the social environments in which technological artifacts are developed, implemented, understood, and adapted. HR scholars have noted that technologies, as productivity enhancing tools, are not ready-made but in the making through complex interactions with organizational and environmental factors. For instance, Passino Jr and Severance (1990) noted that when the adoption of an information technology (IT) system

became a popular innovation activity, global organizations had to deal with the complexity of identifying effective ways to integrate continuously developing IT systems.

HR scholars recognize that the utilization of algorithmic technologies is not a solution ready-made for adoption. It is continuously developed through complex interactions between many stakeholders, including users, developers, deployers, regulators, and the general public (Langer and König 2023). What algorithmic technology means to individuals, and how they respond to it, is influenced by a range of factors within and across organizations. Cao et al. (2023) showed that AI identity, “a subjective AI-related self-view after internalizing the dominant social views on AI-related meanings and values,” affected individuals' ethical behavior. Specifically, individuals with a high level of AI identity might have an elevated view of themselves and a sense of psychological entitlement due to society's high-value viewpoints regarding technology. They suggested this sense of entitlement might lead to unethical behavior. Anicich (2022) found in his autoethnographic field study that app-workers—who appeared to be highly isolated individuals—formulated their identity through social interactions with a broad community. He observed that they liked to share personal stories in online forums in the form of “narrative flexing,” highlighting unusual positive or negative experiences.

As a newly emerged technological artifact, algorithmic management provides opportunities to revisit and revise existing understandings about the process of innovation within and across companies. For instance, Zimmer, Baiyere, and Salmela (2023) proposed a new way to understand technology diffusion through the lens of deinstitutionalization. They offered an insight that the adoption of new technology was not just an activity of “addition” but accompanied the subtraction of existing routines and practices. Prior studies on technology diffusion have emphasized the process of new practice adoption, although many organizations have to undergo a process of abandoning existing practices in order to introduce a new practice. For instance, the introduction of a new digital system may come with the removal of existing formal systems, such as a process of managers' checking and approval. Such “subtraction” activities might entail a sense-making process within the organization about the proper way of working together. Zimmer, Baiyere, and Salmela (2023) argued that the adoption of digital workplace technology comes with not only “addition activities” but also “subtraction activities.”

An emerging concern is to ensure that algorithmic management is employed in a socially desirable way, and that it does not exacerbate inequalities in society. Companies can, with algorithmic technology, easily take advantage of workers who do not have the power to resist the system. Scholars offer normative arguments and proposals on how to make algorithmic systems accountable. For instance, Rahman, Karunakaran, and Cameron (2024) discussed how to make online platform companies accountable. They suggested two views on platform accountability. First, a bottom-up approach that highlights collective actions by less powerful actors, such as gig workers or app developers, to improve rules and resist the dominance of platform owners. This approach tends to emphasize procedural justice. Second, a top-down, institutional approach that aims to prevent exploitative practices and

ensure fairness among various platform participants through legal, regulatory, and governance reforms. The top-down approach mainly focuses on distributive justice. Further, Rahman, Karunakaran, and Cameron (2024) proposed that the two approaches need to be combined to ensure the algorithmic management of platform companies was properly accounted for. Martin (2019) discussed the ethical implications of algorithmic management. He argued that companies must develop algorithms in such a way that organizations take responsibility for the structure of decision delegation within the algorithmic system.

4 | Theme 2: The Algorithmic Technology to Enhance HR Delivery

4.1 | Tool View—Algorithmic Technology as a New HR Tool

When new digital technologies were made available, they were used to enhance the effectiveness of HR functions and activities. For instance, in the 1970s, Bassett (1973) noted that computer technology could enhance “manpower forecasting and scheduling.” The upgrade and innovation of HR practices through technologies continued when HR information systems became popular in the 1990s (Haines and Petit 1997), and internet-based recruiting activities became a norm in the 2000s (Baird, Griffin, and Henderson 2003). This scholarly effort to recognize the benefits of using new technologies for HR delivery has been extended to algorithmic technologies and AI solutions. Scholars began to examine how algorithmic technology could function as a tool to enhance the effectiveness of HR practices. In their systematic literature review, Malik, Budhwar, and Kazmi (2023) noted that AI can indeed “assist” HR decision-makers in delivering the major HR functions, including the acquisition, development, and retention of HRs. Technology vendors provide AI-enabled bots and digitized decision-making solutions that can carry out many HR tasks, enabling organizations to reduce costs and enhance employee experiences (Malik et al. 2022).

The use of algorithmic technologies for HR functions is a relatively new phenomenon, and, therefore, rigorously conducted firm-level field studies are still lacking. However, studies have begun to confirm that algorithmic technologies can improve the delivery of traditional HR functions, bringing benefits to organizations (Meijerink et al. 2021; Prikshat, Malik, and Budhwar 2023). For instance, Hickman et al. (2024) showcased how to use algorithmic technologies to enhance job interview effectiveness, given how accurately ML tools could predict interviewees’ personality traits. Tong et al. (2021) examined how algorithmic technology could be used as a tool to improve performance feedback effectiveness, observing that AI-based performance feedback could enhance that effectiveness. They argued for the advantages of deploying AI rather than human managers for employee feedback, which they termed the “AI deployment effect,” as it quickly analyzed a large amount of performance data with great precision and provided personalized recommendations. Thus, algorithmic performance feedback would enhance employee performance. They also observed some negative effects of algorithmic feedback caused by employees’ negative perceptions of the technology. However, they argued that

those negative perceptions would decline as employees spent more time in the organization. Gaessler and Piezunka (2023) showed that AI systems could be an effective training partner for decision-makers, examining whether algorithmic systems could function as a substitute for human trainers. In their study on chess computers, trainees experienced experiential learning that emulated their experiences with human trainers. Considering the cost and scarcity of human trainers, AI trainers could bring tangible benefits. Nevertheless, algorithmic systems cannot be perfect substitutes for human trainers, as AI trainers do not provide trainees with opportunities to deal with idiosyncratic human errors.

One of the traditional HR activities heavily affected by algorithmic technologies is HR planning and forecasting. The traditional HR activities of workforce planning and forecasting underwent substantial changes and improvements under the rubric of “HR Analytics” (Batistič and Van der Laken 2019; Margherita 2022). *HR Analytics* can be defined as “an HR practice enabled by IT that uses descriptive, visual, and statistical analyses of data related to HR processes, human capital, organizational performance, and external economic benchmarks to establish business impact and enable data-driven decision-making” (Marler and Boudreau 2017). The use of algorithmic technologies in HR analytics dates back to the 1980s when computer technology was popularized (Lawler and Elliot 1996). The advent of AI and ML technologies opened up new possibilities for HR analytics to contribute to organizational performance. Although concrete evidence for the organizational outcomes of HR analytics has yet to be provided, Kryscynski et al. (2018) showed that HR professionals who were more data savvy and had higher analytical capabilities reported higher perceived job performance.

An important stream of research that emerged in relation to algorithms’ impact on HR is job design. Parent-Rocheleau and Parker (2022) summarized that algorithmic technologies seriously affected six traditional managerial functions: monitoring, goal setting, performance management, scheduling, compensation, and job termination. By changing the nature of these functions, algorithms affect job design features, including the degree of autonomy, task variety and significance, job complexity, physical demands, and emotional demands. For instance, algorithms can change an organization’s monitoring practices, thereby making jobs less autonomous and simpler, while increasing emotional demands by tracking workers’ emotional expressions in their jobs. Extending this work, Fraccaroli, Zaniboni, and Truxillo (2024) discussed how algorithmic mechanisms (recommending, restricting, recording, rating, replacing, rewarding) might affect work design features. Furthermore, Bellesia, Mattarelli, and Bertolotti (2023) argued that some of the newly emerged jobs, such as crowd work, could not be fully explained by the existing framework of work design, and proposed a new work characteristic, algorithmic embeddedness, as a crucial defining work design feature of today’s jobs.

Some HR scholars adopt a design science approach, proposing specific algorithmic solutions to HR problems. These studies differ from typical empirical studies, in that they showcase novel methods practitioners can enact to enhance the effectiveness of HR delivery. For instance, Yuan, Kroon, and Kramer (2024)

compared the effectiveness of different predictive models for HR analytics and showed that the random-effects bagged tree produced the most accurate predictions. Chowdhury et al. (2023) argued that in order for an algorithmic HR system to be effective, it should be interpretable in the eyes of HR managers. With this in mind, they showcased the capability of the Local Interpretable Model-Agnostic Explanations technique as an approach to ensure that HR managers could understand and explain the predictions generated by AI-based ML models for a given employee dataset.

4.2 | Proxy View—Employees' Reactions to Algorithmic HR Technology

A new technology-driven HR solution often encounters resistance from employees, though that may be subtle and subjective rather than overt and explicit. This phenomenon is well-documented in earlier studies on HR technological innovations. For instance, in the early 1960s, when electronic data processing technology was applied to personnel management, Munson (1962) noted that introducing a technological innovation in HR was difficult not because the machines were resistant to change but because the humans were. When computer-based performance monitoring systems were introduced, Ball and Wilson (2000) observed that employees were not passive recipients of the new reality. Instead, they used their agency to develop diverse discourses and interpretations of the realities, allowing them to resist change.

With regard to algorithmic HR practices, research has also observed that employees are suspicious of its promise to provide better and fairer outcomes. For instance, Newman, Fast, and Harmon (2020) found that algorithmic HR decisions were perceived less fair than identical human-made decisions, and led to lower organizational commitment due to perceptions of reductionism, that is, algorithms reduce the use of accurate information via quantification and decontextualization. Lavanchy et al. (2023) also found that job applicants thought algorithm-driven recruitment processes were less fair than human-only or algorithm-assisted human processes. Applicants worried that the algorithm would not recognize their uniqueness. Parent-Rocheleau and Parker (2022) examined how algorithmic technology influenced work designs. They concluded that algorithmic HRM is not a predetermined tool. Rather, its impact on work design is moderated by sociotechnical dynamics that involve perceptions of transparency, fairness, and the controllability of the algorithmic systems.

A research theme that has received particular attention is HR professionals' reactions to algorithmic HR technologies, for whom such technologies are sensitive issues, though they may strengthen the professionals' status with analytic capabilities. The technologies may create new HR positions such as HR data analyst. However, algorithmic HR technologies also have the capability to replace their job by automating some decision-making tasks that were previously carried out by HR professionals. Studies suggest that the impact of algorithmic technology on HR's role is not uniform across organizations. Wiblen and Marler (2021) examined how the role of HR professionals was

affected by the adoption of digital technology. They observed interesting variations in the degree to which technology dominated talent management decision-making. While technology dominated talent management in one organization, the existing social relationships prevailed over the technology in another. Leavitt, Barnes, and Shapiro (2024) theorized that algorithmic performance management systems do not automatically eliminate the need for human managers, who could instead develop trust in the system by playing the role of translators and augmenters of algorithms.

Several studies explicitly recognize HR professionals' agentic capability concerning how algorithmic technologies are utilized. Charwood and Guenole (2022) argued that HR professionals have the agency to lead organizations to overcome the risks of bias and unfairness embedded in algorithmic HR technologies. Fu, Keegan, and McCartney (2023) examined HR analysts' storytelling activities in the organization. They observed that HR analysts engaged in two major types of storytelling that seemingly contradicted each other: showcasing and curbing. Showcasing is an effort to gain legitimacy by making other organizational members recognize the value of their work. It raises others' awareness and expectations of HR analytics. Curbing is an effort to reduce enthusiasm for HR analytics in the organization by slowing an HR analytics project or making it less visible therein. McCartney and Fu (2024) confirmed the importance of storytelling for HR professionals, showing that their analytic skills might not lead to job performance without proper storytelling skills.

4.3 | Ensemble View—Social Construction of Algorithmic HR Technologies

HR scholars have recognized that technology-assisted HR cannot be fully understood without considering the formative influences of a range of actors within and outside the organization. For instance, Dineen and Williamson (2012) observed that the design of web-based recruiting practice was shaped by factors outside the organization, such as labor supply, as well as the firm's internal factors, including its compensation practices. Scholars have noted that algorithmic HRM is also subject to the formative influences of many actors within and beyond the organization. Algorithmic HRM systems operate through complex interactions between vendors, users, regulators, and the general public. Langer and König (2023) demonstrated that critical features of algorithmic HRM, such as the opacity of the system, are matters of concern for a range of stakeholders. Myllymäki (2021) argued that digitized HRM was formulated and reproduced through the complex interaction between the agency of many actors and material artifacts.

A distinctive concern regarding algorithmic HR technologies is their potential vulnerability to bias in society (Bogen 2019). Vassilopoulou et al. (2024) explored how biases prevalent in the broader society shaped the way HR managers perceived HR algorithms. They examined how scientism, the unjustified prioritization of the scientific method over other ethical and reasoned arguments, could blind HR professionals and make them uncritically believe HR algorithms function

with scientific impartiality. Kelan (2024) pointed out that AI-supported hiring practices may be affected by biases and prejudices in society against disadvantaged groups (such as females and ethnic minorities). She was concerned that AI-supported hiring is vulnerable to the influence of bias in its data collection process, determining how the data are used, and how the decision is made.

The possibility that algorithmic HR systems may carry biases drove the discussion on regulating AI-assisted HR systems. Dargnies, Hakimov, and Kübler (2024) demonstrated that people's aversion to algorithmic hiring practices was rooted in their concern about gender profiling behind the algorithmic system. Therefore, regulations against biased AI systems are justifiable. Some jurisdictions already have regulations on algorithmic HR systems. An example is New York City Law 2021, regarding Automated Employment Decision Tools, the Bias Audit Law. This law aims to prevent biased decisions by automated HR systems. The European Union is also introducing regulations that provide job seekers with the right not to be solely dictated to by automated decision-making systems (Jibril and Florentina 2024). Hunkenschroer and Luetge (2022) reviewed the literature on AI-enabled recruiting and selection systems and recognized many ethical concerns inherent to AI-enabled recruiting. They advocated even stronger regulations against biased recruitment decisions by AI and recommended employers go beyond current regulations to ensure that predictive AI-enabled hiring tools do not exacerbate the problem of bias in recruitment.

Apart from the legal aspects of algorithmic HR technologies, ethicality attracted attention, too, and substantially so with regard to AI-assisted hiring. Hunkenschroer and Luetge (2022) found 51 articles on this particular topic, despite the relative newness of the phenomenon. Most articles, they found, focused on the algorithmic bias that may affect hiring decisions. Some studies recognized that the ethicality of AI-assisted hiring is socially constructed. For instance, da Motta Veiga, Figueroa-Armijos, and Clark (2023) reported that the ethical perceptions in society about AI-assisted hiring affected organizational attractiveness.

5 | Theme 3: Emergence of New Type of Workers in Algorithmic Management

5.1 | Tool View—Ensuring the Productivity of New Technology Workers

When new technologies arise, new types of workers emerge, necessitating that organizations develop HR solutions to effectively manage them. For instance, in the 1960s, corporations began to hire research scientists for their research and development activities, and then realized these scientists should be managed differently from typical employees (Henry 1962). The advent of algorithmic technology gave rise to several new types of workers, which led to the emergence of new lines of discussion on how to ensure organizations could effectively manage them. In the era of algorithmic management, two new types of workers or work arrangement emerged: platform workers and algorithmic jobs.

Platform workers (or “app-workers”) are those who provide short-term services to clients through online platform companies such as Uber or Amazon (Kuhn and Maleki 2017). These services could entail low-skilled work (e.g., food delivery) or highly technical jobs (e.g., computer programming). Platform workers share some similarities with traditional contractors or temporary workers, as they have limited commitment to the hiring organization. However, they are different to traditional contingent workers, in the sense that their work is almost entirely arranged and monitored by algorithmic systems. Platform workers can provide services even to offshore clients, providing platform companies with opportunities to build a global business (Lehdonvirta et al. 2019). Platform workers have a large degree of flexibility to choose when and for whom to work, although their work experience may not be of much assistance in their long-term career progression (Duggan et al. 2023).

Algorithmic jobs are those that enable organizations to develop, adopt, and utilize algorithmic technologies. Kellogg, Valentine, and Christin (2020) pointed out that algorithmic technologies lead to the emergence of algorithmic occupations. They proposed three types of algorithmic occupation: algorithmic curation, algorithmic brokerage, and algorithmic articulation. The *algorithmic curators* are those involved in generating and preparing data for algorithmic management systems. Organizations could outsource and externalize algorithmic curating work, relying on new types of workers who could be termed “ghost workers” (Gray and Suri 2019), “invisible digi-workers” (Justesen and Plesner 2024), or “micro workers” (Morgan, van Zoonen, and ter Hoeven 2023), and who play a crucial role in developing AI (Altenried 2020). The *algorithmic brokers* help organizational stakeholders understand what the algorithmic systems can do for them. Algorithmic brokers or “translators” (Henke, Levine, and McNerney 2018) help managers and employees make the most of AI technology. Although they are not AI developers, they influence how AI technologies are used, and produce outcomes. Finally, *algorithmic articulators* are those involved in coordinating “who will be doing what, when, where and how, as well as handling missed responsibilities, unfinished jobs and all the steps necessary so that projects do not break down.” Algorithmic articulators include, for instance, project managers and digital consultants who help the organization identify gaps in the use of algorithmic systems, and propose ways to fill them. Another example is the job of checking and authenticating the work of algorithms, regarding, for instance, Uber employees. Its main role is to verify the records of drivers processed in the system.

Both new forms of work—platform work and algorithmic jobs—are proliferating as they provide value to the organization. Since their work arrangements and working environment are quite different to those of traditional employees, different HRM practices and activities are required to ensure their productivity. Managing platform workers might seem to be straightforward, as they are under the control of algorithmic systems. However, ensuring their commitment, and compliance with algorithmic systems, may be more challenging than it appears.

Scholars have examined how platform companies effectively manage and control the app-workers. For instance, Rahman, Weiss, and Karunakaran (2023) examined how platform companies could successfully manage gig workers' sense of autonomy, using "experimental hands." They observed that workers initially experienced increased and then decreased autonomy. Later, workers came to consider diminished autonomy as a normal part of their job. Galière (2020) investigated how food-delivery platform companies managed their platform workers. The findings suggested the company could make the workers believe that the algorithmic reward systems (pay-per-delivery, performance evaluation system, and shift reservation policies) represented a hyper-meritocratic ideal of justice. Zhang et al. (2024) proposed the concept of "perceived algorithmic evaluation" (PAE), which refers to platform workers' perception of their work being evaluated through algorithmic processes, including matching, recording, and rating. They argued that PAE could enhance app-workers' flow experience, and improve their capacity to achieve more in their job. Cameron (2024) examined algorithmic managers manufacturing workplace consent for gig workers. Although algorithmic control tends to create a bad job with a high level of standardization and surveillance, gig workers come to like it, and feel like they have choices under algorithmic management. She observed that gig workers came to believe they were skillful agents who made voluntary choices to work for the company, making the essentially bad job a "good bad" job.

Compared to that of platform workers, research on algorithmic jobs is scant, although several studies offer insights on how organizations can make the most of those jobs. Grönsund and Aanestad (2020) observed that AI technologies brought forth a complex pattern of interplay between humans and algorithms, engendering a "human-in-the-loop" work configuration. In their participant observation study on a business intelligence solution company, the authors found that the introduction of algorithm-based automation necessitated "auditors" to oversee the output from the system, and "data scientists" to analyze the outcome of AI systems and then feed back into the system. The data scientists made adjustments and improvements to the systems as they saw the need. van den Broek, Sergeeva, and Huysman (2021) conducted an ethnographic study on a team that developed an AI-based hiring system. In the process of developing the system, the organization initially excluded the domain experts, aiming to develop algorithms that could do a much better job than human experts. However, the organization eventually undertook a dialectic developmental process, where AI developers worked together with domain experts, creating ML systems that incorporated human expertise. A common suggestion made by these studies was that effective AI systems are likely to emerge when algorithmic systems are designed to incorporate inputs from human experts.

5.2 | Proxy View—New Technology Workers' Agentic Responses to Algorithmic Technology

New types of workers who emerge with new technology tend to have different professional and organizational identities, which imposes new challenges on organizations. When new

technology is introduced, it may affect the established jurisdictions of a profession, making some professionals strive to maintain their autonomy and authority over their jobs. For instance, when research scientists were hired as R&D workers, they demanded a high level of job autonomy and decision-making authority, trying to maintain their professional identity even in the context of organization-based work settings (Bailyn 1985).

Technology professionals' efforts to secure power and autonomy are also observed with regard to algorithmic technologies. A distinctive characteristic of AI technology is its inherent opaqueness and uncertainty that provide both opportunities and threats to professional workers. Waardenburg, Huysman, and Sergeeva (2022) conducted a 31-month ethnographic study on the use of algorithmic technology by the Dutch police, focusing on intelligence officers whose role was to translate ML outcomes to managers of the organization. Interestingly, they found that the opaqueness of ML technology allowed the intelligence officers to be "kins in the land of the blind," sometimes imposing their own judgments on the algorithmic predictions.

AI technology has equivocal and ambiguous features, which enable established professional workers to formulate varying responses to the AI technology that penetrates their work. Einola, Khoreva, and Tienari (2023) conducted a 14-month ethnographic study exploring how organizational members of a media consultancy company extended their collective affects to include advanced technology, when engaging with anthropomorphized virtual AI in the workplace. They observed that while managers talked in abstract terms about how the (ro)bot was supposed to help employees organize their tasks, the employees judged its applicability by the concrete changes it brought to their daily work in the here and now. This rift between the abstractness of transformative strategy, endorsed by management, and the concreteness of operations emanating from different work tasks and responsibilities, highlighted by employees, was an important cause of negative effects and discontent in the organization, which the humanization of AI-fueled. In a similar vein, Scarbrough, Chen, and Patriotta (2024) discovered that radiologists displayed varied responses to AI-based radiology technology, depending on their role and status in the profession. The frontline radiology practitioners perceived AI as a threat that intruded on professional autonomy and interrupted their relationship with patients. In contrast, radiologists who occupied the role of organizational manager tended to perceive AI as a tool to enhance service quality and improve organizational efficiency. Radiologists who considered themselves leaders in the field tended to see AI as a catalyst for change, and sensed AI's potential to further develop the profession.

Studies indicate that also platform workers could express their agency, though their work life is tightly controlled by algorithms. Bellesia, Mattarelli, and Bertolotti (2023) found that knowledge-intensive platform workers exhibited varied behavioral and emotional responses to the algorithmic systems, depending on their perception of algorithms' affordance (such as barriers, individual visibility, self-extension, and rules of the game). Crayne and Brawley Newlin (2024) observed that platform workers' success depended on their self-leadership. Algorithmic management systems were not designed to provide growth opportunities for platform workers. However,

those who had a high level of self-leadership could set their own goals and develop skills, enact their agency, enjoy greater well-being, and increase their chances of moving on to better jobs. However, the authors also noted that enactment of this agency in the platform work environment could be a draining experience for some workers.

5.3 | Ensemble View—Social Construction of New Technology Workers

The meanings technology imparts to workers are influenced by many factors that cut across organizations: a range of occupational, social, cultural, and regulatory environments may work together in shaping the dynamics between technology workers and their organization. HR scholars have recognized this for many decades. For instance, in the 1980s, Gerpott and Domsch (1985) observed the cross-national differences in the management of technology workers. More recently, Adya (2008) observed that IT professionals' career experiences depended on prevailing assumptions in society about gender and ethnic background.

Scholarly efforts to understand the role of broader social, cultural, and institutional environments in shaping the experience of technology workers continued with algorithmic technologies. A key theme that received the most attention was platform workers. For instance, Karanović, Berends, and Engel (2021) examined how platform workers' work experiences were shaped by the regulatory environment. In an empirical study of Uber drivers, they found that platform workers did not always succumb to algorithmic controls. In addition, their responses to algorithmic control varied depending on the regulatory environment in the local area. They compared platform workers' (Uber drivers) responses to algorithmic controls in two different regulatory environments: a direct regulatory environment where regulations were directly imposed on both drivers and the platform company, and an indirect one where regulations were imposed on the platform company, which then enforced the regulations on drivers only. In the direct regulatory environment, the authors observed that platform workers and the platform company tended to develop mutually dependent relationships. In contrast, platform workers in the indirect regulatory environment developed more antagonistic attitudes toward the platform company.

Migrant platform workers are an excellent example of those whose work experiences are shaped by national policies as well as the firm's algorithmic management. In many major cities around the world, platform work is heavily populated by migrant workers (van Doorn, Ferrari, and Graham 2023). It rarely comes with decent working conditions (Alyanak et al. 2023), but does afford the workers crucial opportunities to enhance their livelihood. Further, while algorithmic management systems do not provide migrants with rich career development opportunities, the online platform's algorithmic systems do allow migrants to overcome language barriers. The precarious legal status of migrants allows the company to easily control the workers through an algorithmic system (Schaupp 2022).

Scholars also note that the emergence of platform workers allows us to appreciate companies' dynamic interactions with

their external stakeholders, and revise some of the established assumptions in strategic HRM. For instance, Keegan and Meijerink (2023) pointed out that the growing popularity of online platform companies has allowed us to rethink the established notion that contingent and contract workers provide only tangential benefits to organizations. In the case of platform companies, they argued, contingent workers are an essential part of the company's workforce ecosystem, and sources of competitive advantage. Duggan et al. (2023) found that algorithmic control for app-workers involved not only the organization's algorithmic systems but also non-organizational parties such as suppliers and customers. Meijerink et al. (2024) noted that the gig economy has allowed us to revisit the established conceptions of talent management. Traditionally, talent identification involves intraorganizational decision-makers such as HR professionals and line managers. However, talent management in the context of online labor platforms defies this established conception, as it requires the involvement of not only the company but also the personnel and clients.

The career development of skilled digital workers (such as data scientists and AI developers) is not just a matter of personal choice. It is the outcome of interactions between individual and societal level factors. Castro, van der Heijden, and Henderson (2020) examined the experiences of junior academics in Science, Technology, Engineering, and Mathematics disciplines who switched to working as data scientists. They found that junior researchers' career transition into data science was affected by many factors, including the perceived career uncertainties in academia and the lack of meaning in the professorship. Interestingly, they observed that junior academics' successful transition to data science was facilitated by the re-affirmation of their professional identity as researchers, rather than abandoning academia altogether. Castro, van der Heijden, and Henderson's (2020) research suggested that the successful management of technology workers involved more than promoting their commitment to the current organization. In some countries, developing new technology workers is a matter of national policy. In China, for instance, the government drives national educational systems to develop skilled professionals in AI technology (Hine and Floridi 2024; Roberts et al. 2021).

6 | Future Research Agenda

We encourage future HR researchers examining algorithmic technology to draw inspiration from prior research on HRM and technology in general, and thereby make unique as well as enduring contributions to the literature. Algorithmic technologies provide excellent opportunities to advance our knowledge of HRM-technology relations, not only due to their potential to bring significant changes in HRM, but also because of their unique features that set this technology apart from previous technological innovations. As discussed earlier, algorithmic technology has several distinctive features related to HR, including opacity and nontransparency, the possibility of carrying bias within the system, user aversion to algorithms and preference for human decision-makers, human-likeness that raises the issue of trust and trustworthiness, algorithms as alternative governance mechanisms, and algorithms' capability for decision-making. HR researchers might examine

how such unique features of algorithmic technologies can promote rather than downgrade human potential in organizational settings.

In identifying promising topics for future research, scholars can be guided by the matters of concern for previous researchers on HRM and technology. As reviewed in this article, many scholars have written excellent papers across the major thematic areas of research—technology-driven changes in work structure and design, new HR technologies, and new types of technology workers—and they did so by adopting three approaches to technology, namely the tool, proxy, and ensemble views. We suggest future scientific research on algorithmic technology and HRM should continue expanding these research streams, making enduring contributions drawing on unique features of algorithmic technologies.

Table 2 presents a set of generic research questions that could guide future HR researchers who would like to explore further HRM scholarship on algorithmic technology. This table is not intended to provide a comprehensive list of research questions on algorithmic (or other) technologies. Instead, it aims to help future scholars identify questions with which they can initiate their scientific endeavors regarding the relationships between algorithmic (or other) technologies and HRM.

HR scholars can examine the impact of algorithmic management on work structure and design from several different angles. HR researchers may embrace *the tool view of technology* and examine how algorithmic technology changes the structure and design of work, and thereby enhances firm outcomes. Alternatively, they can explore the patterns of collaboration between algorithms and humans that can produce the best outcomes for the company. For example, Anthony, Bechky, and Fayard (2023) explored the collaboration of AI and humans by explicitly taking the tool view of technology. They recognized that AI technology is different to prior technological tools, with its constant change, invisibility, and inscrutability. Building on these unique features of AI, they proposed a relational approach to studying its impact on work design and structure. Strategic HRM researchers may further extend such views, examining how AI, as a tool, brings out changes to work design and thereby influences a firm's outcomes.

Furthermore, HR researchers could adopt *the proxy view of technology* and examine the sources of algorithm aversion, or the antecedents of algoactivism, or how humans build trust in algorithms in organizational settings. Alternatively, HR scholars could follow *the ensemble view of technology* and unveil how algorithm aversion is affected by the cultural and institutional environment, or how algorithmic transparency is secured by the norms and regulations in a society. In investigating these research questions, it is crucial to look at both the threats and opportunities algorithmic technology enacts on workers. The interdisciplinary perspective that will allow HR researchers to examine reactions from organizational stakeholders beyond individual employees is also important. Doellgast and Wagner (2022) noted that the impact of algorithmic management on work structure and work design attracted increasing attention in labor economics and the sociology of work. HR researchers could learn from neighboring social science disciplines such as this to consider

how the workplace dynamics around algorithmic technology are affected and enacted by factors both within and beyond the organization.

Algorithmic HR technologies should be also further investigated. HR researchers could trace the impact of algorithmic HR technologies by employing *the tool view of technology*. For example, in this special issue, Pei et al. (2024) explored how algorithmic HR technology, with its unique human-like features, could enhance the quality of HR decisions and/or the efficiency of traditional HR activities. Future research could also discuss how algorithmic technology engenders new HR activities. By studying the consequences of the implementation of algorithmic HR technologies in HR practices, HR researchers can examine not only the potential enhancement but also potential perils of using this technology in HR. Future research could also discuss how algorithmic HR technologies augment and superpower some decision-making tasks traditionally performed by HR professionals. *The proxy view of technology* could also be supportive of future research. For example, HR scholars can explore why workers react differently to algorithmic HR technologies. Alternatively, they can follow a social constructionist approach (proxy view), and examine how algorithmic HR technology alters the status and identity of HR professionals. Tandon et al. (2024) examined in this special issue how HR professionals reacted to algorithm-based HR practices, extending the discussion on AI aversion. *The ensemble view of technology* may also inform future HR researchers. For instance, a study could explore how social norms and legal environments shape the effectiveness of algorithmic HR technologies. Future research could further unfold how new regulations on AI influence the development and utilization of algorithmic technologies (Parviainen 2022). These questions also reflect a more in-depth examination of how and why algorithmic HR technologies impact organizations and employees, as identified over time in our review.

Third, HR scholars can examine the emergence of platform workers and algorithmic jobs from different angles. Equipped with *the tool view of technology*, scholars could study how algorithm workers (curators, brokers, articulators) contribute to the success of an organization. Furthermore, they could investigate how organizations can attract and retain high-quality platform workers, and maintain their motivation to serve the interests of the company. *The proxy view of technology* can also be helpful in investigating this thematic area. HR researchers could investigate how algorithm workers formulate their professional identities (Kellogg, Valentine, and Christin 2020). Alternatively, they can explore the drivers of platform workers' satisfaction and commitment in various life domains. Researchers can also employ the *ensemble view of technology*, and delve into longitudinal or ethnographic research designs investigating how algorithm workers' careers are shaped by cultural and institutional environments, or how migrant platform workers make sense of their work. In line with our review, these questions reflect an inclination toward paying more attention to the unique identity and needs of new types of workers, and the contributions they can make to organizations. Researchers can also reflect on the broader social and cultural implications of algorithm workers. For instance, HR researchers could build on Burrell and Fourcade's (2021)

comment on the emergence of the power imbalance between the “coding elites,” who have power over the digital means of production, and “cybertariats” who contribute low-paid labor to this new economy.

7 | Discussion

This special issue makes timely contributions to advancing the literature on the role of algorithmic technology in HRM. Five papers collectively exemplify the research that builds on and extends the broad scholarly conversations on HRM-technology relations, focusing on the unique benefits and challenges algorithmic technology brings to HRM. Specifically, Pei et al. (2024) promote the idea that technologies can be a tool to enhance the effectiveness of HR practices. The authors examine how the negative effects of negative feedback can be moderated by utilizing an AI-based feedback system. They found that AI-based feedback systems can function as a “remediation” tool that mitigates employees’ fear of receiving negative feedback. While this study is in line with the previous studies that advocated the value of new HR technology as a tool to enhance firm productivity, it advances the broader literature on HRM-technology relations by highlighting the human-likeness of AI technology as an agent of feedback provider. Edwards et al. (2024) extend the research stream on the impact of new HR technology on employees. This study investigates when and how algorithmic HRM can enhance employee motivations, and, in this sense, follows the research tradition that adopts the tool view of technology. However, the authors explicitly acknowledge the agency of employees who formulate their own responses to new technology, as prior scholars did with the proxy view of the technology. Specifically, the authors found that employees’ reactions to an automated HR metric system are shaped by their belief about management’s intentions behind the system. When employees believe their organization utilizes an algorithmic HR system as a device for employee control, they experience a high level of extrinsic motivation, primarily focusing on metricized tasks while neglecting those that are not. When employees believe their organization uses an algorithmic system to provide constructive feedback, they exhibit less emotional exhaustion and more intrinsic motivation. This study adds value to the broader conversation on HRM-technology relations, highlighting that what might matter more to employees is their perception of *why* new technology-driven HR is implemented, rather than *which* technology is used.

Lu et al.’s (2024) paper is in line with the tradition of taking the proxy view of the technology, examining how employees react to new technology-driven algorithmic controls. The empirical study is set in the context of food-delivery platform work in China. The algorithmic management system in platform companies automates the entire HR process, that is, recruiting workers, allocating tasks, evaluating work performance, and providing payments, setting this technology substantially apart from previous technological innovations adopted in HR. The authors found that algorithmic control negatively affects employee engagement. However, the negative impact of algorithmic control weakens when workers have a higher workload. This may sound counterintuitive as employees would not welcome a high load of repetitive work. The study advances the knowledge on

HRM-technology relations by suggesting that algorithmic technology may provide employees with qualitatively different experiences to previous technologies. Tandon et al. (2024) address one of the most frequently examined topics in HRM-technology research, that is, users’ acceptance of technology. This research stream typically draws on the proxy view of technology, as it highlights users’ agency to perceive and react to the new technology. This study found that HR professionals are ambivalent about algorithmic technology: while they are aversive to this powerful technology, they also appreciate what it can do for them. The authors’ work contributes to the broader discussion on HRM-technology relations, highlighting the human agency of HR professionals who are at the forefront of HR technology adoption. Lamers, Meijerink, and Rettagliata (2024) question paradigmatic ideologies in the scholarship on algorithmic management. Their study builds on the proxy and ensemble views of technology, highlighting the human agency in the development and utilization of algorithmic technology. In this conceptual work, the authors identify two ontological assumptions underlying algorithmic management: the ontology of algorithms and the ontology of management. They argue that both ontological assumptions have their own “blind spots” to overcome. The study advances the broad research on HRM and technology by inviting scholars to explicitly reflect on the underlined ontological assumptions with which researchers and practitioners understand new technology, sometimes without realizing they are employing such assumptions.

Our review and the papers in this special issue provide many practical implications for managers working with algorithmic technologies. To begin, managers must have a clear understanding of how new technologies, such as AI, ML, and NLP, are fundamentally changing work structures, HR processes, and employee roles. Algorithmic tools change the way we work and, thereby, alter the design of jobs. Algorithmic instruments boost the efficiency of HR delivery by streamlining HR procedures and facilitating complex decision-making, but they also raise issues around employee trust, transparency, and algorithmic bias. Algorithmics creates new algorithmic jobs that require special HR attention. Managers should balance algorithm use with human oversight, fostering trust and addressing ethical concerns about fairness. Importantly, these technologies can augment rather than replace human labor, particularly in tasks requiring creativity and strategic insight. This calls for novel techniques to manage employees who collaborate with or are supervised by algorithms, along with strategies to reduce resistance and support adoption. Understanding the larger institutional and cultural factors that influence employee views of these technologies is also crucial for their successful integration into HR practices.

The practical insights from this special issue highlight several enduring themes related to integrating algorithmic systems into HRM. One important message is the need to understand the impact of algorithmic HR systems on employee well-being. Edwards et al. (2024) concern that although algorithms can improve performance via minimal oversight, they may also lead to emotional fatigue and exhaustion. They warn that “the potential harm of misdirected implementation of these systems should not be ignored” (p. 707). Lu et al. (2024) note that excessive algorithmic management may negatively impact employees’

well-being. Lamers, Meijerink, and Rettagliata (2024) warn of the danger of a control-focused “homo economicus” view of algorithmic management. HR managers need to acknowledge that algorithmic HR systems, despite their promise of efficiency and effectiveness, may negatively affect employee's experiences and undermine business outcomes. Therefore, caution is advised regarding excessive dependence on AI, with recommendations for leaders to foster inclusive environments that encourage employees' voice behavior and welcome their engagement in the design and implementation processes, thereby enhancing system acceptability. In line with this, Lu et al. (2024) recommend companies establish effective feedback mechanisms and provide training to help employees deal with the stress of algorithmic management.

Another important insight is that the effectiveness of algorithmic HR systems is contingent on various factors. For instance, Pei et al. (2024) suggest that, in the context of face culture (such as that of China); AI could be more effective than humans in delivering negative feedback. Tandon et al. (2024) suggest that the effective development and use of algorithmic HR systems may depend on employees' prior domain experience and expertise. Edwards et al. (2024) highlight the importance of recognizing employees' HR attributions as moderators of the effectiveness of algorithmic HR systems. These findings suggest that, while algorithmic systems have the potential to improve HRM, their effectiveness is dependent on careful implementation, clear communication, and the incorporation of human-centered approaches that take into account the specific contexts in which algorithmic HR systems are being deployed.

8 | Conclusions

The use of algorithmic technology has dramatically transformed how companies manage their work structure, HR processes are organized, and the new technology is engendering new types of workers. Our review highlights that HR researchers, including the authors of the papers in this special issue, recognize the uniqueness of algorithmic technologies, while building on the long tradition of scholarly work on HRM-technology relations. We also observe that researchers recognize both the potential and limitations of these technologies, noting that while they can improve efficiency, reduce biases, and enable data-driven decision-making, they also raise concerns about transparency, trust, and ethical use. As algorithmic management becomes more prevalent, addressing issues such as bias, transparency, and accountability is essential to fostering trust. Therefore, we call in particular for a balanced and nuanced approach to algorithmic HRM, urging both scholars and practitioners to create inclusive and ethical applications that harness the power of technology while safeguarding human values.

Acknowledgments

The authors thank Co-Editors-in-Chief Howard Klein and Fang Lee Cooke for their support and guidance throughout the work. The authors would also like to thank Mikko Vesa for the inspiration and support provided. This article was financially supported by Marcus Wallenbergs Stiftelse för Internationellt Vetenskapligt Samarbete (Decision of 19.12.2022).

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The authors have nothing to report.

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